Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed.

Supervised learning uses labeled training data to learn a mapping from inputs to outputs. Common examples include classification and regression.

Unsupervised learning finds hidden patterns in data without labeled responses. Clustering and dimensionality reduction are typical tasks.

Reinforcement learning trains an agent to make decisions by rewarding desired behaviors and punishing undesired ones.

A neural network is a computational model inspired by the human brain, consisting of layers of interconnected nodes called neurons.

Deep learning uses neural networks with many layers to learn representations of data with multiple levels of abstraction.

Gradient descent is an optimization algorithm that iteratively adjusts model parameters to minimize a loss function.

Overfitting occurs when a model learns the training data too well including its noise and performs poorly on unseen data.

Retrieval-Augmented Generation combines a retrieval system with a generative model to ground responses in external knowledge.

ChromaDB is an open-source vector database designed for storing and querying embeddings efficiently with support for persistence.

A vector embedding is a numerical representation of text in a high-dimensional space where similar items are placed closer together.

Prompt engineering is the practice of crafting input prompts to guide language models toward producing desired outputs.

Hallucination in LLMs refers to the generation of plausible-sounding but factually incorrect or unsupported information.

Fine-tuning adapts a pre-trained model to a specific task by continuing training on a smaller task-specific dataset.

BERT is a transformer-based model trained on masked language modeling and next sentence prediction tasks.

GPT is an autoregressive language model that generates text by predicting the next token in a sequence.

Cosine similarity measures the angle between two vectors and is commonly used to compare embeddings in semantic search.

A large language model is trained on massive text corpora and can perform a wide variety of language tasks via prompting.

Regularization techniques such as L1 L2 and dropout help prevent overfitting by penalizing model complexity.

The transformer architecture uses self-attention mechanisms to process sequences in parallel and is widely used in NLP tasks.